

d-CCSSBU L and S Charts and Poisson u Chart Code and Their Outputs for Equipment A and B

1 Read Excel Data

Each Excel Data contains three spreadsheet: DAR, CAR, and Mic. Each spreadsheet should have variables Event, Location, Result, USL, LOD, LOQ (For Mic, it does not have LOD and LOQ). You need to change the following path to yours and download these data, code to your local system. You need to have R/RStudio with necessary R packages installed.

You can also realize the algorithms in the manuscript directly in other programming languages.

2 d-CCSSBU-L Function

The function `d_ccssbu_l()` uses Excel data as above to generate D-CCSSBU-L chart.

3 Selection of d for d-CCSSBU-L Function

- `d_ccssbu_l()` constructs the d-CCSSBU-L chart for a specified value of d .
- `d_ccssbu_l_d_selection_arl1()` applies `d_ccssbu_l()` across all candidate values $d = 1, \dots, q$ using Excel input data and summarizes OOC counts and ARL_1 performance for DAR, CAR, and Mic residue types.
- `make_d_ccssbu_l_arl1_wide_tbl()` converts the output of `d_ccssbu_l_d_selection_arl1()` into a wide-format summary table.
- `render_d_ccssbu_l_arl1_tbl()` renders the output table from `make_d_ccssbu_l_arl1_wide_tbl()` as a responsive tibble that automatically adjusts to page width.

- `render_d_ccssbu_l_recommendation_tbl()` generates the recommended d selection table for each residue type (DAR, CAR, and Mic).

4 Application of the functions to Equipment A and B.

4.1 d-CCSSBU-L Selection of d and ARL1 Results for Equipment A

Applying the `d_ccssbu_l_d_selection_arl1()` function in [1] to Equipment A data produces a list of candidate values of d for each residue type. For a given residue type (e.g., DAR), the output contains q rows, where q is the number of ordered clusters identified.

For each candidate value of d , the first d clusters are selected to form the calibration dataset containing n_{Cal} cleaning events. The calibration dataset is then used in the stratified smoothed bootstrap procedure to estimate the chart UCL.

For the resulting UCL, the numbers of OOC events among ND events (OO_{C_ND}), Stage 3A events (OO_{C_3A}), and Stage 3B events (OO_{C_3B}) are reported. An $OO_{C_ND} > 0$ indicates that at least one ND event is identified as OOC. Because ND events represent the best-quality cleaning events, such a result suggests that the calibration dataset may not be representative of the IC process. Similarly, $OO_{C_3A} > 0$ may indicate that the selected calibration dataset does not adequately represent an IC Stage 3A process.

For comparison with conventional control charts, the sampling-location size associated with each cleaning event is recorded, and the median value is used as the benchmark subgroup size. This benchmark subgroup size is used to construct the conventional $\bar{X}$ chart and S chart and to calculate the corresponding benchmark $\bar{X}$ -chart ARL_1 values reported in the table.

Finally, the d-CCSSBU-L chart ARL_1 values are calculated for a range of shift magnitudes. For the d-CCSSBU-L chart, the shift size is defined as $\delta_k = k \times UCL/3$, where k denotes the shift multiplier. The benchmark $\bar{X}$ -chart ARL_1 values are provided for comparison with the proposed chart.

Table 1: d-CCSSBU-L ARL1 simulation results for Equipment A. For each d , the d-CCSSBU-L chart was fitted using the location-specific calibration algorithm. ARL1 was estimated by shifting the bootstrap location statistics using delta $k(d)$ equal to k times $UCL_{boot}(d)$ divided by 3.

Residue	d	q	case_label	nCal	OO _C _ND	OO _C _3A	OO _C _3B	benchmark_n	Xbar_chart_ARL1_k_0p5	UCL	ARL1_k_0p5	ARL1_k_1	ARL1_k_1p5	ARL1_k_2	ARL1_k_2p5	ARL1_k_3
CAR	1	1	Case 1	10	0	0	0	3	60.8923	4.1222	36.5631	36.5631	36.5631	35.9712	1.0000	1.0000
DAR	1	3	Case 2	3	0	0	0	6	26.3658	3.9132	6.8097	3.7064	2.2026	1.4708	1.0641	1.0000
DAR	2	3	Case 2	6	0	0	0	6	26.3658	4.5849	21.5285	7.2046	3.4764	2.0640	1.3702	1.0000
DAR	3	3	Case 2	10	0	0	0	6	26.3658	4.0912	50.7614	28.9855	9.1996	5.8428	2.5445	1.0002
Mic	1	1	Case 1	10	0	0	0	2	91.5158	2.0736	9.9751	9.9751	9.9751	9.9751	9.9751	1.2997

The recommendation algorithm recommended CAR $d=1$, DAR $d = 2$ by Case 1 in Step 8 although it totally has 3 clusters, and Mic $d = 1$.

Table 2: Recommended d values for the d-CCSSBU-L chart for Equipment A.

Residue	recommended_d	q	Xbar_chart_ARL1_k_0p5	D1_values	D2_values	D3_values	D4_values	selected_case	selection_rule	D_star_values	selection_path	recommendation_notes
CAR	1	1	60.8923	1	1	1		Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	
DAR	2	3	26.3658	1, 2, 3	1, 2, 3	1, 2	3	Case 1	max(D1 intersect D2 intersect D3)	1, 2	Case 1: max(D1 intersect D2 intersect D3)	
Mic	1	1	91.5158	1	1	1		Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	

4.2 d-CCSSBU-L Selection of d and ARL1 Results for Equipment B

The `d_ccssbu_l_d_selection_arl1()` function was applied to the Equipment B data using the same procedure described for Equipment A.

For CAR, the candidate solution with $d = 1$ produced OOC ND events (`OOC_ND` = 3). Although all OOC observations were ND, the ordered-cluster algorithm identified two distinct groups of ND events, corresponding to relatively smaller and larger ND values. When only the first cluster was selected for calibration, the calibration dataset became dominated by the smaller ND values, resulting in an excessively small UCL. Consequently, several events containing larger ND values were identified as OOC.

This result illustrates that a calibration dataset should be representative of the IC process, even when all observations are ND. Therefore, the candidate solution with $d = 1$ was not considered suitable or the calibration dataset is not representative.

Table 3: d-CCSSBU-L ARL1 simulation results for Equipment B. For each d , the d-CCSSBU-L chart was fitted using the location-specific calibration algorithm. ARL1 was estimated by shifting the bootstrap location statistics using delta $k(d)$ equal to k times $\text{UCLboot}(d)$ divided by 3.

Residue	d	q	case_label	nCal	OOC_ND	OOC_3A	OOC_3B	benchmark_n	Xbar_chart_ARL1_k_0p5	UCL	ARL1_k_0p5	ARL1_k_1	ARL1_k_1p5	ARL1_k_2	ARL1_k_2p5	ARL1_k_3
CAR	1	2	Case 4	10	3	3	6	3	60.8923	2.1310	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
CAR	2	2	Case 4	20	0	0	0	3	60.8923	30.9387	500.0000	9.9701	6.3837	4.6083	3.5549	1.0000
DAR	1	26	Case 4	10	0	0	0	5	33.4221	19.5308	1.0088	1.0088	1.0088	1.0088	1.0088	1.0000
DAR	2	26	Case 5	13	0	0	0	5	33.4221	19.5454	1.2732	1.0051	1.0035	1.0032	1.0029	1.0000
DAR	3	26	Case 5	23	0	0	0	5	33.4221	23.4040	17.0648	3.4626	2.0206	1.3437	1.0264	1.0008
DAR	4	26	Case 5	29	0	0	0	5	33.4221	20.7417	6.8399	3.9131	2.9291	1.9916	1.0162	1.0000
DAR	5	26	Case 5	39	0	0	0	5	33.4221	20.7252	19.0476	7.7942	4.6992	2.7941	1.1529	1.0004
DAR	6	26	Case 5	49	0	0	0	5	33.4221	20.4053	31.4961	12.5156	7.5472	4.0875	1.3605	1.0014
DAR	7	26	Case 5	59	0	0	0	5	33.4221	19.9548	44.0529	16.5289	11.4548	6.8236	1.7751	1.0003
DAR	8	26	Case 5	69	0	0	0	5	33.4221	19.6397	64.7249	22.3714	17.2117	10.2564	2.4393	1.0001
DAR	9	26	Case 5	79	0	0	0	5	33.4221	19.5022	90.4977	30.3030	24.8756	14.8478	3.2321	1.0000
DAR	10	26	Case 5	89	12	9	0	5	33.4221	19.3354	133.3333	39.1389	34.5423	19.5122	4.2256	1.0001
DAR	11	26	Case 5	99	12	9	0	5	33.4221	19.2718	198.0198	51.9481	46.8384	27.1739	5.3262	1.0001
DAR	12	26	Case 5	109	12	9	0	5	33.4221	19.1111	312.5000	83.6820	76.3359	38.4615	8.2781	1.0009
DAR	13	26	Case 5	117	12	9	0	5	33.4221	18.9497	434.7826	97.5610	91.3242	42.2833	9.5283	1.0014
DAR	14	26	Case 5	124	12	9	0	5	33.4221	18.9689	500.0000	124.2236	118.3432	51.8135	10.8342	1.0010
DAR	15	26	Case 5	126	12	9	0	5	33.4221	19.0971	434.7826	119.0476	109.2896	60.9756	11.7716	1.0012
DAR	16	26	Case 5	136	12	9	0	5	33.4221	18.6654	666.6667	136.9863	135.1351	59.1716	13.7741	1.0011
DAR	17	26	Case 5	140	12	9	0	5	33.4221	19.1061	500.0000	144.9275	139.8601	77.5194	15.3139	1.0008
DAR	18	26	Case 5	150	12	9	0	5	33.4221	18.4172	666.6667	188.6792	186.9159	66.2252	17.0358	1.0013
DAR	19	26	Case 5	156	12	9	0	5	33.4221	14.2788	206.1856	206.1856	186.9159	37.3832	9.2166	1.0019
DAR	20	26	Case 5	162	12	9	0	5	33.4221	14.1476	210.5263	208.3333	194.1748	41.6667	10.6838	1.0023
DAR	21	26	Case 5	165	12	9	0	5	33.4221	16.2715	307.6923	202.0202	190.4762	44.4444	15.5280	1.0030
DAR	22	26	Case 5	175	15	10	0	5	33.4221	13.7396	289.8551	285.7143	263.1579	50.2513	10.9769	1.0036
DAR	23	26	Case 5	185	15	10	0	5	33.4221	13.9869	289.8551	289.8551	277.7778	53.0504	12.5000	1.0038
DAR	24	26	Case 5	190	15	10	0	5	33.4221	13.9279	307.6923	307.6923	289.8551	61.1621	14.0647	1.0000
DAR	25	26	Case 5	199	15	10	0	5	33.4221	13.8748	317.4603	307.6923	60.7903	14.7820	1.0000	
DAR	26	26	Case 5	206	15	10	0	5	33.4221	13.6825	434.7826	425.5319	400.0000	76.9231	16.0256	1.0091
Mic	1	1	Case 4	10	0	0	0	3	60.8923	4.0806	28.6944	28.6944	28.6944	28.6944	28.6944	1.7109

The following table summarizes the recommended values of d for Equipment B. For CAR and Mic, only one admissible candidate remained after applying the selection rules, resulting in recommended values of $d = 2$ and $d = 1$, respectively. For DAR, multiple admissible candidates were identified,

and the selection procedure recommended $d = 6$, corresponding to the smallest ARL_1 among the admissible candidates.

Table 4: Recommended d values for the d-CCSSBU-L chart for Equipment B.

Residue	recommended_d	q	Xbar_chart_ARL1_k_0p5	D1_values	D2_values	D3_values	D4_values	selected_case	selection_rule	D_star_values	selection_path	recommendation_notes
CAR	2	2	60.8923	2	2	1	2	Case 6	min(D1 intersect D2 intersect D4)	2	Case 6: min(D1 intersect D2 intersect D4)	
DAR	6	26	33.4221	1, 2, 3, 4, 5, 6, 7, 8, 9	1, 2, 3, 4, 5, 6, 7, 8, 9	1, 2, 3, 4, 5, 6	7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26	Case 1	max(D1 intersect D2 intersect D3)	1, 2, 3, 4, 5, 6	Case 1: max(D1 intersect D2 intersect D3)	
Mic	1	1	60.8923	1	1	1		Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	

5 d-CCSSBU-S Function

The function `d_ccssbu_s()` uses Excel data as above to generate D-CCSSBU-S chart.

6 Selection of d for d-CCSSBU-S Function

- `d_ccssbu_s()` constructs the d-CCSSBU-L chart for a specified value of d .
- `d_ccssbu_s_d_selection_arl1()` applies `d_ccssbu_s()` across all candidate values $d = 1, \dots, q$ using Excel input data and summarizes OOC counts and ARL_1 performance for DAR, CAR, and Mic residue types.
- `make_d_ccssbu_s_arl1_wide_tbl()` converts the output of `d_ccssbu_s_d_selection_arl1()` into a wide-format summary table.
- `render_d_ccssbu_s_arl1_tbl()` renders the output table from `make_d_ccssbu_s_arl1_wide_tbl()` as a responsive tibble that automatically adjusts to page width.
- `render_d_ccssbu_s_recommendation_tbl()` generates the recommended d selection table for each residue type (DAR, CAR, and Mic).

7 Application of the d Selection Functions to Equipment A and B.

7.1 d-CCSSBU-S Selection of d and ARL_1 Results for Equipment A

Applying the `d_ccssbu_s_d_selection_arl1()` function in [1] to the Equipment A data produces candidate values of d for each residue type. The d-selection procedure is the same as that described for the d-CCSSBU-L chart, except that the scale-specific calibration algorithm is used. For each candidate value of d , the corresponding OOC counts, UCL, benchmark subgroup size, benchmark S -chart ARL_1 , and d-CCSSBU-S chart ARL_1 values are reported.

Table 5: d-CCSSBU-S ARL1 simulation results for Equipment A. For each d, the d-CCSSBU-S chart was fitted using the scale-specific calibration algorithm. ARL1 was estimated by shifting the bootstrap scale statistics using delta k(d) equal to k times UCLboot(d) divided by 3.

Residue	d	q	case_label	nCal	OOB_ND	OOB_3A	OOB_3B	benchmark_n	S_chart_ARL1_k_0p5	UCL	ARL1_k_0p5	ARL1_k_1	ARL1_k_1p5	ARL1_k_2	ARL1_k_2p5	ARL1_k_3
CAR	1	1	Case 1	10	0	0	0	3	98.2929	0.2011	206.1856	63.8978	22.8050	8.6393	2.8736	1.0000
DAR	1	3	Case 2	3	0	0	0	6	58.1871	4.6356	44.8430	18.9753	10.6496	3.3784	2.6438	1.0000
DAR	2	3	Case 2	7	0	0	0	6	58.1871	3.3661	540.5405	112.3596	29.8063	24.6305	10.4221	1.0003
DAR	3	3	Case 2	10	0	1	0	6	58.1871	3.1582	285.7143	87.3362	10.6952	6.5402	3.0492	1.0002
Mic	1	1	Case 1	10	0	0	0	2	132.1139	2.0516	9.9751	9.9751	9.9751	9.9751	9.9751	1.2997

The following Table 6 summarizes the recommended values of d for the d-CCSSBU-S chart for Equipment A. The recommended value was $d = 1$ for CAR, DAR, and Mic. For DAR and Mic, the recommended candidate achieved a smaller $ARL_1(0.5)$ than the corresponding benchmark S chart (Table 5).

For CAR, the recommended candidate did not achieve a smaller $ARL_1(0.5)$ than the benchmark S chart. This result is attributable to the predominance of ND events in the calibration dataset, which produced a very small bootstrap scale statistic and, consequently, a very small UCL. Because the shift magnitude is defined as $\delta_k = k \times UCL/3$, the resulting shift at $k = 0.5$ is also very small. Consequently, the simulated OOB distribution differs only slightly from the IC distribution, making scale shifts difficult to detect and resulting in a large $ARL_1(0.5)$.

Table 6: Recommended d values for the d-CCSSBU-S chart for Equipment A.

Residue	recommended_d	q	S_chart_ARL1_k_0p5	D1_values	D2_values	D3_values	D4_values	selected_case	selection_rule	D_star_values	selection_path	recommendation_notes
CAR	1	1	98.2929	1	1		1	Case 2	min(D1 intersect D2 intersect D4)	1	Case 2: min(D1 intersect D2 intersect D4)	D3 empty: no candidate d satisfied $ARL_1(d, 0.5)$ less than or equal to the conventional S-chart benchmark.
DAR	1	3	58.1871	1, 2, 3	1, 2	1	2, 3	Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	
Mic	1	1	132.1139	1	1	1		Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	

7.2 d-CCSSBU-S Selection of d and ARL1 Results for Equipment B

Table 7: d-CCSSBU-S ARL1 simulation results for Equipment B. For each d, the d-CCSSBU-S chart was fitted using the scale-specific calibration algorithm. ARL1 was estimated by shifting the bootstrap scale statistics using delta k(d) equal to k times UCLboot(d) divided by 3.

Residue	d	q	case_label	nCal	OOC_ND	OOC_3A	OOC_3B	benchmark_n	S_chart_ARL1_k_0p5	UCL	ARL1_k_0p5	ARL1_k_1	ARL1_k_1p5	ARL1_k_2	ARL1_k_2p5	ARL1_k_3
CAR	1	2	Case 4	10	0	0	0	3	98.2929	16.7963	96.1538	23.1750	7.5103	3.3881	1.7827	1.0092
CAR	2	2	Case 4	20	0	0	0	3	98.2929	14.2480	109.2896	22.3214	10.1626	6.3735	3.9331	1.0000
DAR	1	26	Case 4	10	0	1	2	5	67.2648	0.6525	181.8182	68.0272	24.3605	6.6028	1.7899	1.0000
DAR	2	26	Case 4	20	0	1	0	5	67.2648	19.1504	39.7614	39.7614	39.7614	29.4118	29.4118	1.0003
DAR	3	26	Case 5	23	0	1	0	5	67.2648	19.0995	63.2911	32.7332	30.4414	24.1255	6.8611	1.0004
DAR	4	26	Case 5	33	0	1	0	5	67.2648	18.8003	111.7318	62.1118	51.8135	37.6648	12.1507	1.0021
DAR	5	26	Case 5	41	0	1	0	5	67.2648	18.6429	180.1802	90.9091	84.7458	63.6943	19.1571	1.0005
DAR	6	26	Case 5	51	0	1	0	5	67.2648	18.2699	322.5806	183.4862	141.8440	93.0233	30.5344	1.0018
DAR	7	26	Case 5	60	0	1	0	5	67.2648	17.5787	476.1905	208.3333	175.4386	101.5228	38.0228	1.0054
DAR	8	26	Case 5	63	0	1	0	5	67.2648	18.3464	312.5000	200.0000	168.0672	125.7862	42.6439	1.0053
DAR	9	26	Case 5	73	0	1	0	5	67.2648	13.7839	298.5075	285.7143	232.5581	63.8978	39.7614	1.0107
DAR	10	26	Case 5	83	0	1	0	5	67.2648	13.4980	317.4603	298.5075	208.3333	16.2075	11.0988	1.0120
DAR	11	26	Case 5	89	0	1	0	5	67.2648	12.7163	588.2353	500.0000	134.2282	17.9372	7.0721	1.0128
DAR	12	26	Case 5	99	0	1	0	5	67.2648	12.6304	555.5556	434.7826	156.2500	22.6244	8.8928	1.0146
DAR	13	26	Case 5	101	0	1	0	5	67.2648	7.3237	96.1538	24.9688	18.5529	11.1545	4.2526	1.0115
DAR	14	26	Case 5	111	0	1	0	5	67.2648	7.6154	259.7403	33.7268	25.0313	14.0252	5.0994	1.0122
DAR	15	26	Case 5	121	0	1	0	5	67.2648	10.5916	645.1613	434.7826	44.9438	25.1572	6.6979	1.0160
DAR	16	26	Case 5	128	0	1	0	5	67.2648	11.4061	666.6667	555.5556	66.8896	30.6748	8.5288	1.0219
DAR	17	26	Case 5	138	0	1	0	5	67.2648	7.2456	131.5789	47.9616	35.0877	17.1233	4.7382	1.0147
DAR	18	26	Case 5	148	0	1	0	5	67.2648	7.7705	400.0000	63.4921	47.3934	24.4798	5.9737	1.0000
DAR	19	26	Case 5	153	0	1	0	5	67.2648	6.5651	81.9672	61.5385	39.7614	16.5700	4.9751	1.0113
DAR	20	26	Case 5	159	0	1	0	5	67.2648	8.4309	588.2353	84.0336	56.8182	31.1042	7.4267	1.0113
DAR	21	26	Case 5	169	0	1	0	5	67.2648	6.5147	81.9672	63.8978	44.4444	19.4742	5.8038	1.0090
DAR	22	26	Case 5	179	0	1	0	5	67.2648	6.6566	133.3333	89.2857	59.5238	25.5754	7.1048	1.0088
DAR	23	26	Case 5	183	0	1	0	5	67.2648	6.3784	99.0099	80.0000	54.4959	24.0674	6.7889	1.0083
DAR	24	26	Case 5	189	0	1	0	5	67.2648	6.3870	122.6994	93.8967	59.3472	25.5428	7.1124	1.0000
DAR	25	26	Case 5	196	0	1	0	5	67.2648	6.4983	123.4568	94.3396	62.3053	29.6736	8.1800	1.0059
DAR	26	26	Case 5	206	0	1	0	5	67.2648	6.3374	142.8571	108.1081	73.8007	32.2581	7.6423	1.0000
Mic	1	1	Case 3	10	0	0	0	3	98.2929	3.9834	132.4503	132.4503	132.4503	132.4503	132.4503	1.8406

Table 8: Recommended d values for the d-CCSSBU-S chart for Equipment B.

Residue	recommended_d	q	S_chart_ARL1_k_0p5	D1_values	D2_values	D3_values	subclust_cases	subclust_rules	D1_rule_values	subclust_rules	recommendation_rules
CAR	2	98.2929	1,2	1	1	2	Case 1	mean(D1) intersect (D2 intersect D3)	1	Case 1: mean(D1) intersect (D2 intersect D3)	D1 empty: all candidates d values flagged as false since essential Stage 3A event as OOC.
DAR	2	67.2648	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26	1	2,3	1,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26	Case 10	mean(D1) intersect (D2)	2,3	Case 10: mean(D1) intersect (D2)	D1 empty: no candidate d certified ARL1.0, 0.5% less than or equal to the conventional S-chart benchmark.
Mic	1	98.2929	1	1	1	1	Case 2	mean(D1) intersect (D2 intersect D3)	1	Case 2: mean(D1) intersect (D2 intersect D3)	

8 d-CCSSBU-L And d-CCSSBU-S Charts of Equipment A and B

8.1 d-CCSSBU Charts for Equipment A and B with Recommended d Values

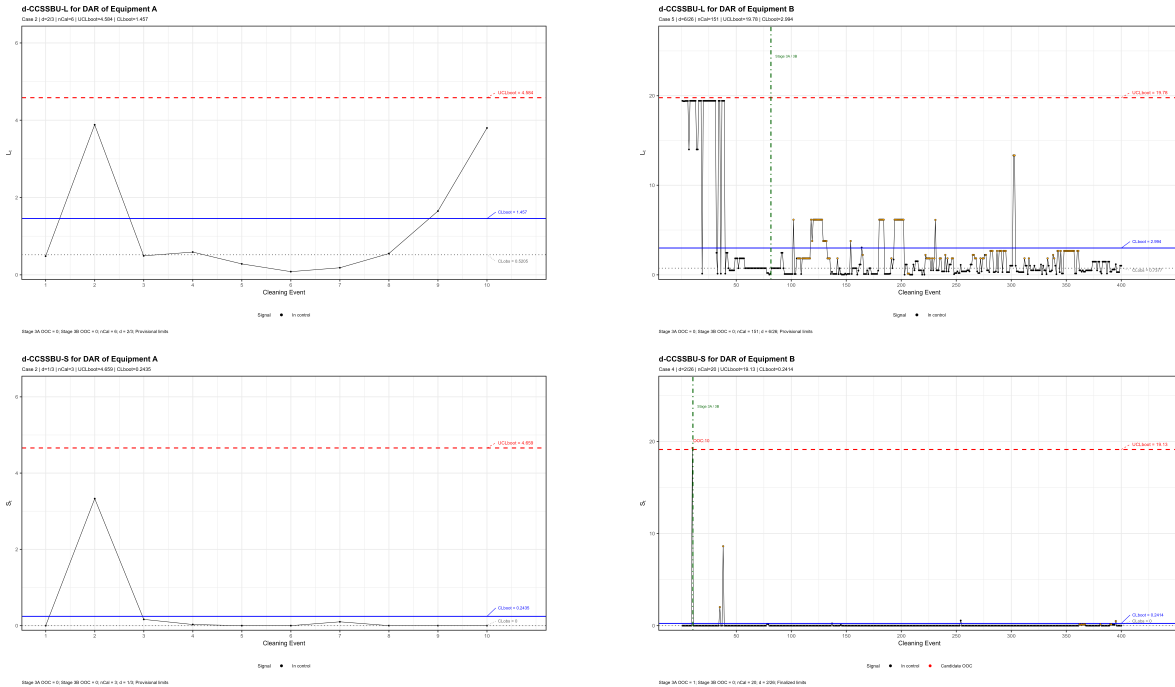
The d-CCSSBU-L and d-CCSSBU-S charts for DAR using the recommended values of d for Equipment A and Equipment B. Similar to the conventional $\bar{X}$ -chart and S -chart pair, the d-CCSSBU-L chart monitors process location, whereas the d-CCSSBU-S chart monitors within-event variability. Together, the two charts provide complementary assessments of process stability.

Equipment A corresponds to Stage 3A data and Equipment B corresponds to Stage 3B data. Previous analyses showed that conventional $\bar{X}$ - S charts and median-MAD charts may generate a large number of OOC signals for heterogeneous cleaning validation data [1]. In contrast, the d-CCSSBU charts provide stable monitoring performance while accounting for data heterogeneity.

The proposed charts are calibrated using d highest-risk ordered clusters. The underlying principle is that if the highest-risk group remains stable and in control, then lower-risk groups are also expected

to remain in control. This risk-based calibration strategy enables a common control limit to be established for heterogeneous cleaning validation data.

For Equipment B, no OOC signals were observed on the d-CCSSBU-L chart, indicating stable process location performance. On the d-CCSSBU-S chart, Event 10 was identified as OOC. Inspection of the corresponding DAR observations (15.93, 0.11, 0.11, 15.93, and 32.95) shows substantial within-event variability, resulting in a large MAD value. The d-CCSSBU-S chart therefore identified this event as exhibiting unusually high within-event variation.

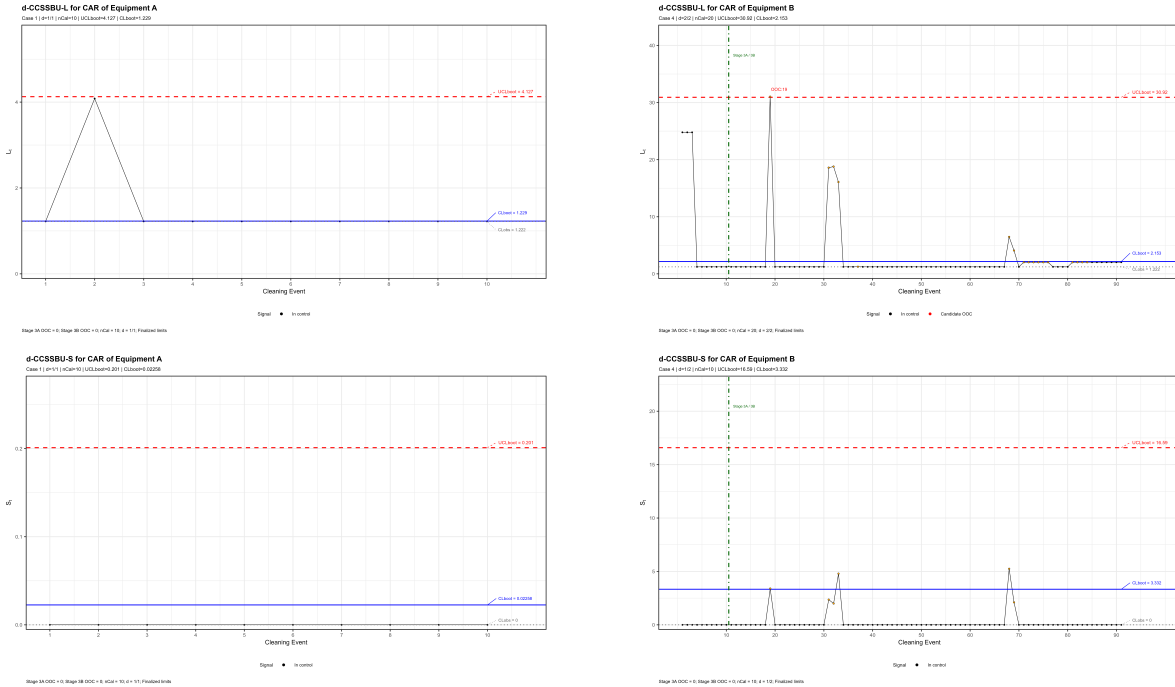


The following charts present the d-CCSSBU-L and d-CCSSBU-S charts for CAR using the recommended values of d for Equipment A and Equipment B. Similar to the conventional $\bar{X}$ - S chart pair, the d-CCSSBU-L chart monitors process location, whereas the d-CCSSBU-S chart monitors within-event variability.

CAR data typically form a single cluster because most observations are ND or close to ND. For Equipment B, two CAR groups corresponding to the original cleaning process and the improved cleaning process were intentionally included to demonstrate the capability of the proposed framework to evaluate process changes. The d-CCSSBU charts show that the improved cleaning process using the new cleaning agent remains comparable with the original cleaning process and can be assessed using a common control limit, despite the process change. Consistent with expectations, most CAR observations in both groups were ND events.

For Equipment B, Event 19 was identified as an OOC signal by the d-CCSSBU-L chart but not by the d-CCSSBU-S chart. The corresponding CAR observations were 35.49, 31.99, and 24.80, yielding a median value of 31.99, which was substantially higher than those of the surrounding events. However, the within-event variability remained relatively small, as reflected by the modest MAD value. Consequently, the event was detected as a location shift rather than a scale shift.

This example illustrates the complementary roles of the two charts. The d-CCSSBU-L chart is sensitive to changes in process level, while the d-CCSSBU-S chart is sensitive to changes in within-event variability. Together, they provide a comprehensive assessment of cleaning process stability.

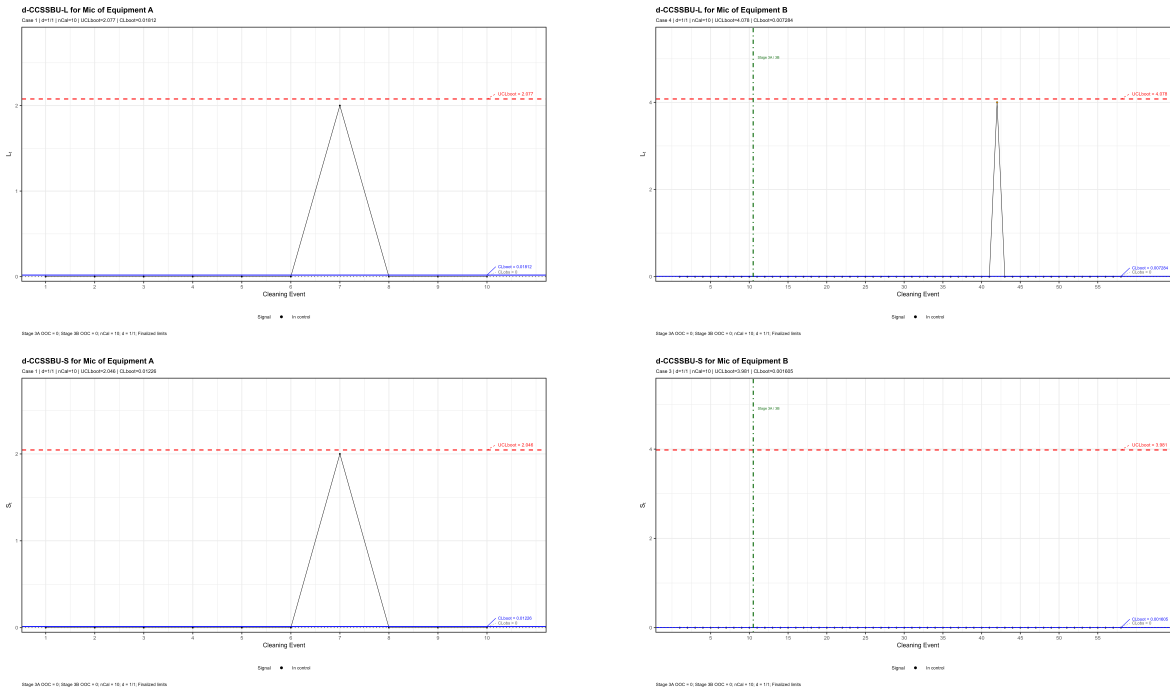


The following figure presents the d-CCSSBU-L and d-CCSSBU-S charts for Mic using the recommended values of d for Equipment A and Equipment B. Because microbial count data are homogeneous and contain predominantly zero counts, both the location and scale statistics remain near zero for most cleaning events.

For Equipment A, Event 7 contains microbial counts of 0 and 1. Consequently, both the event location statistic and within-event variability increase relative to the remaining events, producing a visible spike on both the d-CCSSBU-L and d-CCSSBU-S charts. All other events contain zero counts and therefore have location and scale statistics close to zero.

For Equipment B, Event 42 contains microbial counts of 1 and 1. As a result, the event location statistic is elevated relative to the surrounding events, producing a spike on the d-CCSSBU-L chart. However, because both observations are identical, the MAD is zero and no corresponding increase is observed on the d-CCSSBU-S chart.

These examples illustrate the complementary roles of the two charts. The d-CCSSBU-L chart detects shifts in event location, whereas the d-CCSSBU-S chart detects changes in within-event variability. For homogeneous microbial count data with predominantly zero counts, both charts correctly reflect the underlying process characteristics.



9 Poisson u Chart Illustration for Mic Data

- `prepare_mic_data()` prepares Mic data.
- `poisson_gof_test()` tests if Mic data follows Poisson distribution.
- `poisson_overdispersion_test()` tests if Mic data is over-dispersed.
- `poisson_mic_u_chart()` draw Poisson u-chart. It uses above three functions.

9.0.1 Mic of Equipment A

The following table shows that the Mic data for Equipment A satisfy the Poisson assumptions, including the goodness-of-fit test and overdispersion assessment. Therefore, a conventional Poisson u chart can be applied to this dataset.

Figure X compares the Poisson u chart with the d-CCSSBU-L and d-CCSSBU-S charts for the same data. The Poisson u chart and the d-CCSSBU-L chart produce nearly identical results, with UCL values of 2.0974 and 2.07, respectively. Both charts identify the same process behavior and no OOC events are observed.

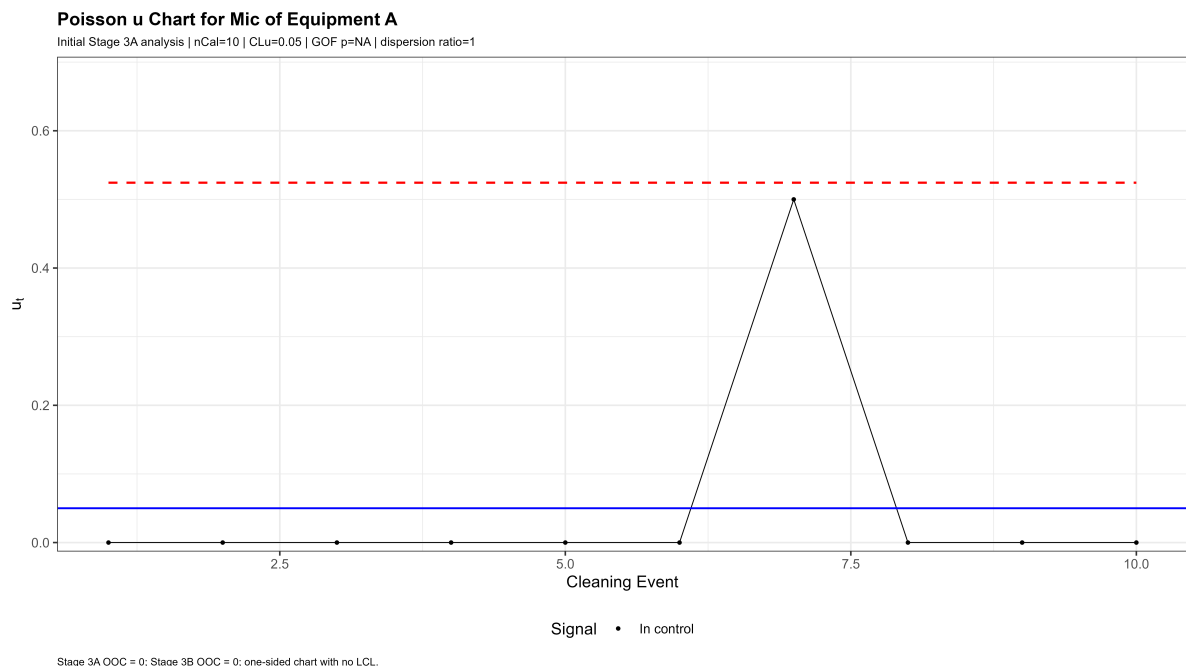
The close agreement is expected because the Mic data are homogeneous, consist predominantly of zero counts, and satisfy the Poisson distribution assumption. Under these conditions, the d-CCSSBU-L chart effectively reproduces the performance of the conventional Poisson u chart.

For the same reason, the d-CCSSBU-L and d-CCSSBU-S charts are also highly similar. Most cleaning events contain only zero counts, resulting in both the location and scale statistics remaining

close to zero throughout the monitoring period. Consequently, the location and scale charts exhibit nearly identical patterns.

Table 9: Mic data Poisson assessment and chart selection for Equipment A.

Equipment	Case	USL	GOF p	GOF pass	Dispersion	Overdispersion pass	Mic Poisson	CL	UCL min	UCL max	Stage 3A OOC	Stage 3B OOC
Equipment A	Case 3	25	0.627	TRUE	1	TRUE	TRUE	0.2	2.0974	2.0974	0	0



9.0.2 Mic of Equipment B

The following table shows that the Mic data for Equipment B do not satisfy the Poisson assumptions. The goodness-of-fit test failed ($p = 0.0274$), and the overdispersion assessment also failed with a dispersion ratio of 1.6286. Consequently, the Mic data were not considered Poisson, and a conventional Poisson u chart was not appropriate for this dataset.

This conclusion is also supported by the d-CCSSBU charts. Unlike Equipment A, where the d-CCSSBU-L and d-CCSSBU-S charts were nearly identical and closely matched the Poisson u chart, the location and scale charts for Equipment B exhibit different patterns. The d-CCSSBU-L chart reflects differences in event-level microbial counts, whereas the d-CCSSBU-S chart reflects within-event variability. The visual differences between the two charts indicate that the underlying process behavior is not consistent with a simple Poisson mechanism, for which the mean and variance are intrinsically linked.

The chart results therefore agree with the formal statistical assessment. Both the goodness-of-fit and overdispersion tests indicate that the Poisson assumption is inadequate for the Equipment B Mic data, while the differing behaviors of the d-CCSSBU-L and d-CCSSBU-S charts provide a visual confirmation of this conclusion. Nevertheless, no OOC events were identified for either Stage 3A

190 or Stage 3B, indicating that the microbial monitoring process remained stable throughout the study
 191 period.

Table 10: Mic data Poisson assessment and chart selection for Equipment B.

Equipment	Case	USL	GOF p	GOF pass	Dispersion	Overdispersion pass	Mic Poisson	CL	UCL min	UCL max	Stage 3A OOC	Stage 3B OOC
Equipment B	Case 3	25	0.0274	FALSE	1.6286	FALSE	FALSE	0.3333	1.488	2.3333	0	0

192 1. Yang CX. D-CCSSBU-l-or-s control chart for cleaning validation [Internet]. GitHub repository; 2026. Available from: <https://github.com/ChandlerXiandeYang/d-CCSSBU-L-or-S-Control-Chart>